

Operating Systems for Intelligent Systems (ENGS2306)

Assignment 3

User & Group Administration Solution

Presented to : **Eng. Mohammed AL-Hams**

Student name: **Deema Mohammed AL-Maqadma**

Student number: : **2320230766**

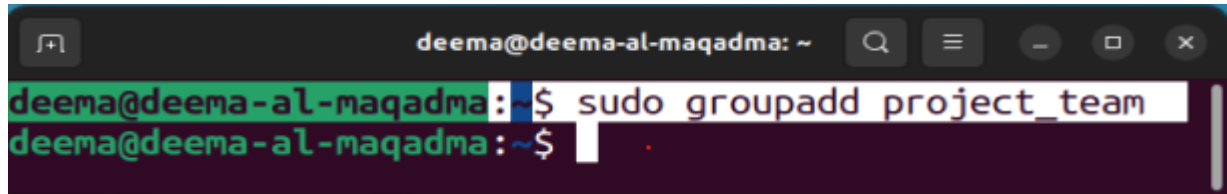
Semester: Second Semester 2024/2025

1. Create a new group named project_team

sudo groupadd project_team

Explanation:

The groupadd command creates a new group. We need sudo because creating groups requires administrative privileges.

A terminal window titled 'deema@deema-al-maqadma: ~' showing the command 'sudo groupadd project_team' being entered and executed. The prompt changes from 'deema@deema-al-maqadma:~\$' to 'deema@deema-al-maqadma:~\$' after the command is run.

```
deema@deema-al-maqadma:~$ sudo groupadd project_team
deema@deema-al-maqadma:~$
```

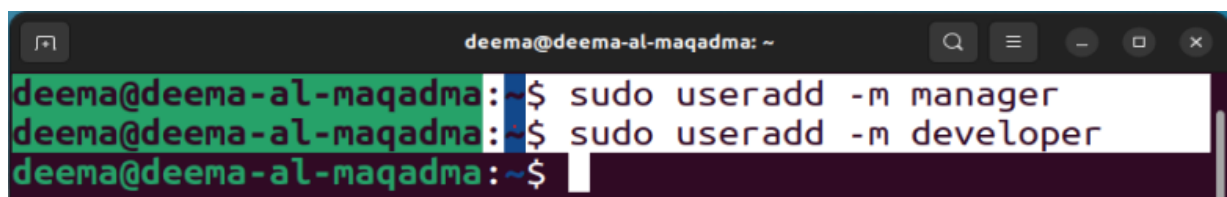
2. Create two new users with home directories

sudo useradd -m manager

sudo useradd -m developer

Explanation:

useradd creates new user accounts. The -m flag ensures the home directory is created for each user.

A terminal window titled 'deema@deema-al-maqadma: ~' showing two commands being executed: 'sudo useradd -m manager' and 'sudo useradd -m developer'. The prompt changes from 'deema@deema-al-maqadma:~\$' to 'deema@deema-al-maqadma:~\$' after each command is run.

```
deema@deema-al-maqadma:~$ sudo useradd -m manager
deema@deema-al-maqadma:~$ sudo useradd -m developer
deema@deema-al-maqadma:~$
```

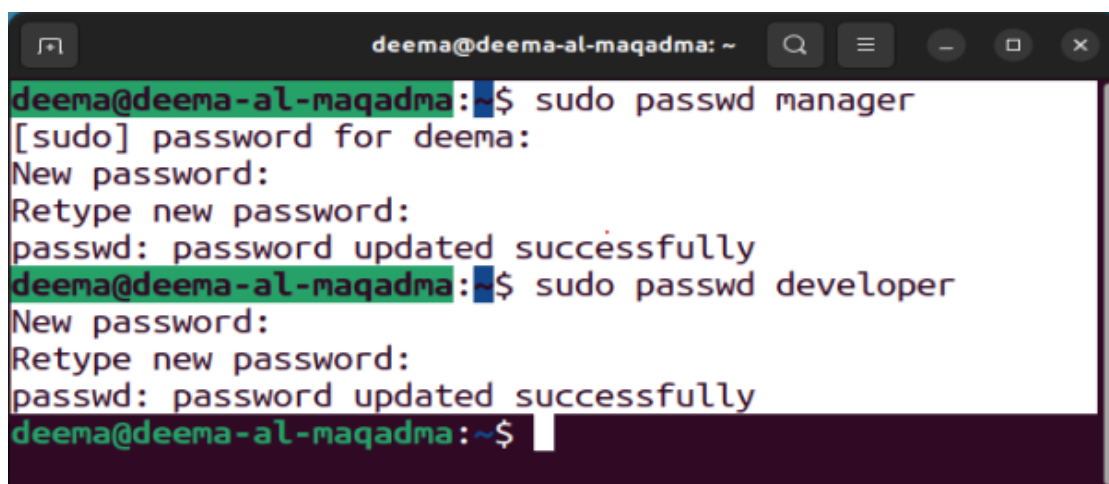
3. Set simple passwords for both users

sudo passwd manager # "manag_er123"

sudo passwd developer # "develop_er123"

Explanation:

The passwd command changes a user's password. For this exercise, we're using simple passwords.

A terminal window titled 'deema@deema-al-maqadma: ~' showing two password change commands. For 'sudo passwd manager', it prompts for the current password (which is empty), then for a new password and its retype, resulting in 'passwd: password updated successfully'. The same process is repeated for 'sudo passwd developer'. The prompt changes from 'deema@deema-al-maqadma:~\$' to 'deema@deema-al-maqadma:~\$' after each command is run.

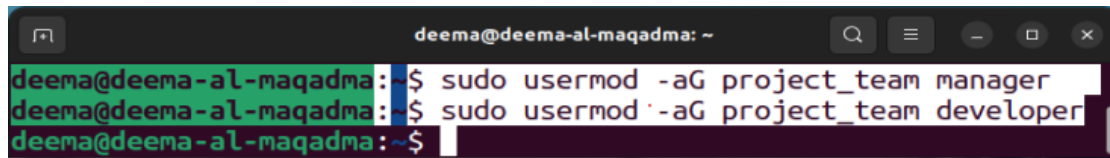
```
deema@deema-al-maqadma:~$ sudo passwd manager
[sudo] password for deema:
New password:
Retype new password:
passwd: password updated successfully
deema@deema-al-maqadma:~$ sudo passwd developer
New password:
Retype new password:
passwd: password updated successfully
deema@deema-al-maqadma:~$
```

4. Add users to the project_team group

sudo usermod -aG project_team manager
sudo usermod -aG project_team developer

Explanation:

usermod modifies user accounts. -aG adds the user to supplementary groups without removing them from other groups.

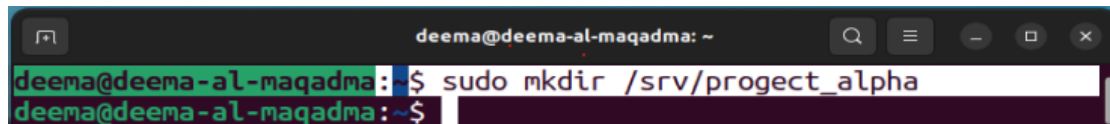
A terminal window titled 'deema@deema-al-maqadma: ~' showing two commands being executed. The first command is 'sudo usermod -aG project_team manager' and the second is 'sudo usermod -aG project_team developer'. Both commands are preceded by the prompt 'deema@deema-al-maqadma:~\$' and followed by a new prompt line 'deema@deema-al-maqadma:~\$'.

5. Create the project directory

sudo mkdir /srv/project_alpha

Explanation:

mkdir creates a new directory. /srv is a standard location for site-specific data.

A terminal window titled 'deema@deema-al-maqadma: ~' showing the command 'sudo mkdir /srv/project_alpha' being executed. The prompt is 'deema@deema-al-maqadma:~\$' and the command is followed by a new prompt line 'deema@deema-al-maqadma:~\$'.

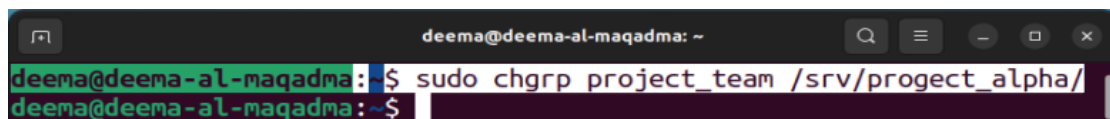
6. Change directory group ownership

sudo chgrp project_team /srv/project_alpha

OR // sudo chown root:project_team /srv/project_alpha

Explanation:

chgrp changes the group ownership of files/directories. Here we set it to our project team.

A terminal window titled 'deema@deema-al-maqadma: ~' showing the command 'sudo chgrp project_team /srv/project_alpha/' being executed. The prompt is 'deema@deema-al-maqadma:~\$' and the command is followed by a new prompt line 'deema@deema-al-maqadma:~\$'.

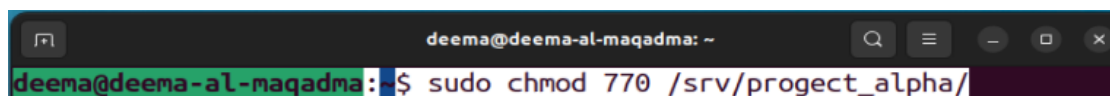
7. Set directory permissions

sudo chmod 770 /srv/project_alpha

Explanation:

chmod changes file permissions. 770 means:

- Owner (root): read, write, execute (7)
- Group (project_team): read, write, execute (7)
- Others: no permissions (0)

A terminal window titled 'deema@deema-al-maqadma: ~' showing the command 'sudo chmod 770 /srv/project_alpha/' being executed. The prompt is 'deema@deema-al-maqadma:~\$' and the command is followed by a new prompt line 'deema@deema-al-maqadma:~\$'.

8. Verification

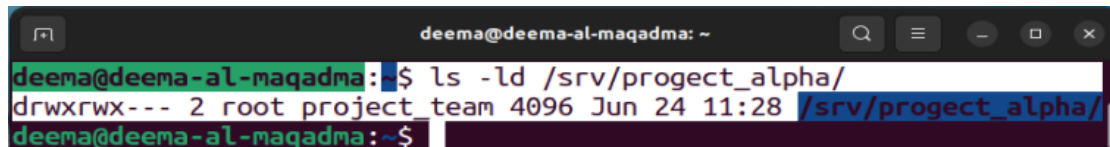
ls -ld /srv/project_alpha

Expected Output:

Explanation:

The ls -ld command shows directory permissions and ownership. The output confirms:

- drwxrwx--- matches our 770 permissions
- Owned by root with group project_team



```
deema@deema-al-maqadma: ~  
deema@deema-al-maqadma:~$ ls -ld /srv/project_alpha/  
drwxrwx--- 2 root project_team 4096 Jun 24 11:28 /srv/project_alpha/  
deema@deema-al-maqadma:~$
```

9. Report Explanation :

English Explanation:

Using a group to manage access to /srv/project_alpha provides scalability and easier administration. When new team members join, we simply add them to the project_team group rather than modifying each file's permissions individually. This approach maintains consistent permissions across all project resources and reduces the risk of permission errors. Group-based permissions also make it easier to audit who has access to project files.

"واخر دعواهم ان الحمد لله رب العالمين".